1. Executive KPI Query

```sql
SELECT
    ROUND(SUM(Revenue),2) AS Total_Revenue,

    COUNT(DISTINCT VisitorID) AS Total_Visitors,

    COUNT(DISTINCT OrderID) AS Total_Orders,
```

2.

```sql
    ROUND(
        SUM(PurchasedFlag) * 100.0 /
        NULLIF(SUM(ProductViewedFlag),0),
        2
    ) AS Conversion_Rate_Percent,
```

3.

```sql
    ROUND(
        SUM(CartAbandonmentFlag) * 100.0 /
        NULLIF(SUM(AddedToCartFlag),0),
        2
    ) AS Cart_Abandonment_Rate_Percent,
```

4.

```sql
    ROUND(
        SUM(Revenue) /
        NULLIF(COUNT(DISTINCT OrderID),0),
        2
    ) AS Average_Order_Value,
```

5.

```sql
    ROUND(
        SUM(MarketingCost) /
        NULLIF(SUM(PurchasedFlag),0),
        2
    ) AS Customer_Acquisition_Cost,
```

6.

7.

```
    ROUND(
        SUM(Revenue) /
        NULLIF(SUM(MarketingCost),0),
        2
    ) AS ROAS
```

8.

```
FROM ecommerce_dataset;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Total_Revenue	Total_Visitors	Total_Orders	Conversion_Rate_Percent	Cart_Abandonment_Rate_Percent	Average_Order_Value	Customer_Acquisition_Cost	ROAS
464881	2300	1503	29.99	43.91	309.30	199.11	3.38

2. Chart Query — Revenue Trend Analysis

Power BI Chart

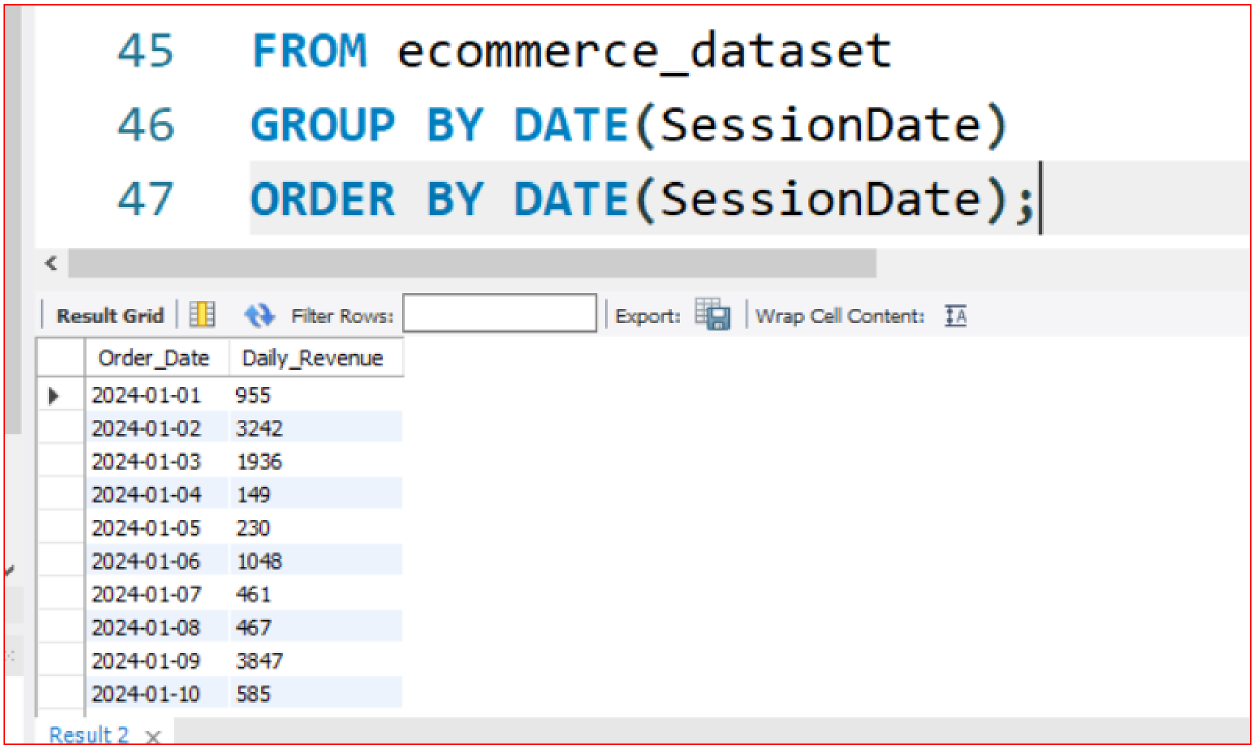
Line Chart

Axis

SessionDate

Values

Total Revenue



```
FROM ecommerce_dataset
GROUP BY DATE(SessionDate)
ORDER BY DATE(SessionDate);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Order_Date	Daily_Revenue
2024-01-01	955
2024-01-02	3242
2024-01-03	1936
2024-01-04	149
2024-01-05	230
2024-01-06	1048
2024-01-07	461
2024-01-08	467
2024-01-09	3847
2024-01-10	585

Result 2

Insight

Shows revenue growth patterns, seasonal demand, and sales fluctuations.

3. Chart Query — Conversion Funnel Analysis

Power BI Chart

Funnel Chart

```
SELECT
'Product Viewed' AS Funnel_Stage,
SUM(ProductViewedFlag) AS Total_Customers

FROM ecommerce_dataset

UNION

SELECT
'Added To Cart',
```

SUM(AddedToCartFlag)

FROM ecommerce_dataset

UNION

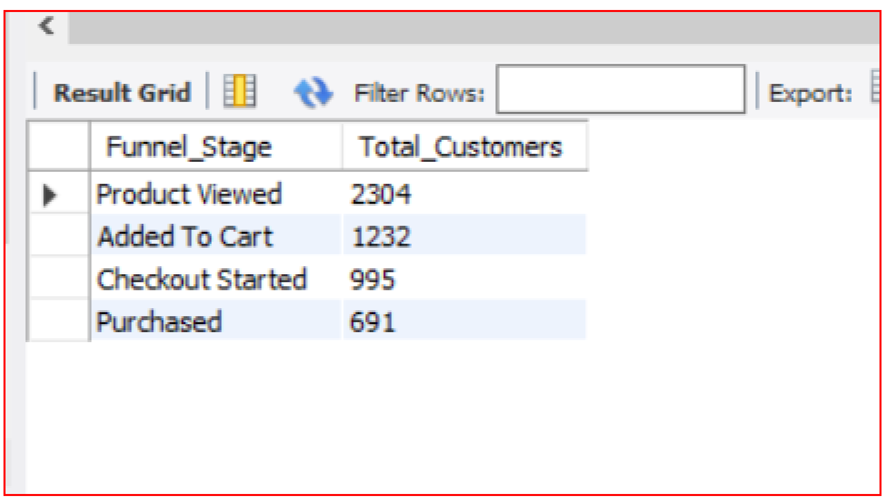
SELECT
'Checkout Started',
SUM(CheckoutStartedFlag)

FROM ecommerce_dataset

UNION

SELECT
'Purchased',
SUM(PurchasedFlag)

FROM ecommerce_dataset;



Funnel_Stage	Total_Customers
Product Viewed	2304
Added To Cart	1232
Checkout Started	995
Purchased	691

Insight

Identifies customer drop-off points during the purchase journey.

4. Chart Query — Cart Abandonment by Traffic Channel

Power BI Chart

Clustered Bar Chart

Axis

TrafficChannel

Values

Cart Abandonment Rate

```
SELECT
TrafficChannel,

ROUND(
SUM(CartAbandonmentFlag) * 100.0 /
NULLIF(SUM(AddedToCartFlag),0),
2
) AS Cart_Abandonment_Rate

FROM ecommerce_dataset
GROUP BY TrafficChannel
ORDER BY Cart_Abandonment_Rate DESC;
```

Result Grid | Filter Rows: | Exp

TrafficChannel	Cart_Abandonment_Rate
Email	48.59
Social Media	48.37
Affiliate	45.91
Influencer	44.85
Paid Search	42.95
Organic Search	42.50
Referral	40.35
Direct	38.89

Insight

Reveals low-quality traffic channels causing purchase abandonment.

5. Chart Query — Marketing ROI by Channel

Power BI Chart

Clustered Column Chart

```
SELECT
TrafficChannel,

ROUND(SUM(Revenue),2) AS Revenue,

ROUND(SUM(MarketingCost),2) AS Marketing_Cost,

ROUND(
SUM(Revenue) /
NULLIF(SUM(MarketingCost),0),
2
) AS ROAS
```

```
FROM ecommerce_dataset
GROUP BY TrafficChannel
ORDER BY Revenue DESC;
```

TrafficChannel	Revenue	Marketing_Cost	ROAS
Direct	85270	17508.42	4.87
Referral	74838	18771.58	3.99
Social Media	57907	15942.03	3.63
Paid Search	57700	16674.09	3.46
Affiliate	56822	17188.53	3.31
Organic Search	46447	19372.08	2.4
Email	45707	15501.46	2.95
Influencer	40190	16629.6	2.42

Insight

Compares revenue generation vs marketing spend effectiveness.

6. Chart Query — Device Conversion Performance

Power BI Chart

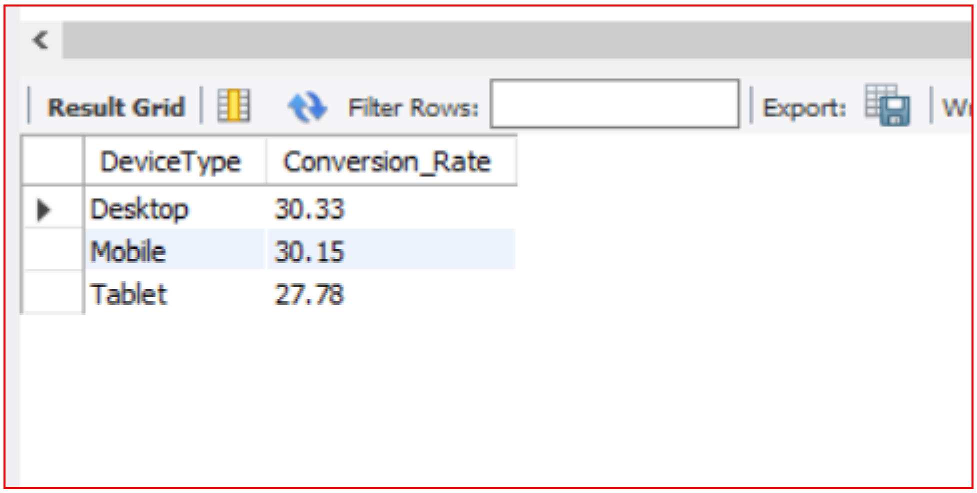
Donut Chart

```
SELECT
DeviceType,

ROUND(
SUM(PurchasedFlag) * 100.0 /
NULLIF(SUM(ProductViewedFlag),0),
2
```

) AS Conversion_Rate

FROM ecommerce_dataset
GROUP BY DeviceType
ORDER BY Conversion_Rate DESC;



Result Grid | Filter Rows: | Export:

DeviceType	Conversion_Rate
Desktop	30.33
Mobile	30.15
Tablet	27.78

Insight

Shows which device drives highest purchase completion.

7. Chart Query — Customer Segment Revenue Contribution

Power BI Chart

Treemap

SELECT
CustomerSegment,

ROUND(SUM(Revenue),2) AS Revenue

FROM ecommerce_dataset

GROUP BY CustomerSegment
ORDER BY Revenue DESC;

CustomerSegment	Revenue
High Value	124149
Returning	107343
Discount Shopper	83827
New	80434
VIP	69128

Insight

Highlights the most profitable customer segments.

8. Chart Query — Regional Marketing Performance

Power BI Chart

Map Chart

SELECT
Region,

ROUND(SUM(Revenue),2) AS Revenue,

ROUND(
SUM(Revenue) /
NULLIF(SUM(MarketingCost),0),
2
) AS ROAS

FROM ecommerce_dataset
GROUP BY Region
ORDER BY Revenue DESC;

Region	Revenue	ROAS
South America	86139	3.62
Africa	85374	3.86
Europe	78359	3.26
Asia Pacific	77727	3.56
Middle East	77382	3.49
North America	59900	2.53

Insight

Shows geographic sales and marketing performance.

9. Chart Query — Revenue by Marketing Channel

Power BI Chart

Stacked Bar Chart

SELECT
TrafficChannel,
ROUND(SUM(Revenue),2) AS Revenue

FROM ecommerce_dataset
GROUP BY TrafficChannel
ORDER BY Revenue DESC;

TrafficChannel	Revenue
Direct	85270
Referral	74838
Social Media	57907
Paid Search	57700
Affiliate	56822
Organic Search	46447
Email	45707
Influencer	40190

Insight

Identifies strongest acquisition channels.

10. Chart Query — Customer Engagement Analysis

Power BI Chart

Scatter Chart

X-Axis

Pages Viewed

Y-Axis

Revenue

```
SELECT
PagesViewed,
ROUND(AVG(Revenue),2) AS Avg_Revenue

FROM ecommerce_dataset
GROUP BY PagesViewed
ORDER BY PagesViewed;
```

PagesViewed	Avg_Revenue
1	261.08
2	225.07
3	168.29
4	215.89
5	202.40
6	350.33
7	181.45
8	125.08
9	187.08
10	206.44

Insight

Shows relationship between engagement and spending.

11. Advanced KPI Query — Monthly Performance Dashboard

```
SELECT
YEAR(SessionDate) AS Year,

MONTH(SessionDate) AS Month,

ROUND(SUM(Revenue),2) AS Revenue,

COUNT(DISTINCT VisitorID) AS Visitors,

COUNT(DISTINCT OrderID) AS Orders,

ROUND(
SUM(PurchasedFlag) * 100.0 /
NULLIF(SUM(ProductViewedFlag),0),
```

```sql
2
) AS Conversion_Rate,

ROUND(
SUM(CartAbandonmentFlag) * 100.0 /
NULLIF(SUM(AddedToCartFlag),0),
2
) AS Cart_Abandonment_Rate,

ROUND(
SUM(Revenue) /
NULLIF(SUM(MarketingCost),0),
2
) AS ROAS

FROM ecommerce_dataset
GROUP BY
YEAR(SessionDate),
MONTH(SessionDate)

ORDER BY Year, Month;
```

Year	Month	Revenue	Visitors	Orders	Conversion_Rate	Cart_Abandonment_Rate	ROAS
2024	1	20180	147	93	29.93	45.68	2.22
2024	2	32487	152	94	35.53	37.93	3.44
2024	3	35226	144	92	35.42	40.70	4.35
2024	4	23423	138	81	28.26	45.83	2.72
2024	5	30815	155	110	29.03	36.62	2.99
2024	6	34997	123	81	30.89	34.48	4.83
2024	7	45942	135	95	41.48	36.36	5.94
2024	8	27094	153	92	32.68	35.90	2.99
2024	9	40674	130	88	30.77	46.67	5.48
2024	10	33113	141	91	26.24	51.95	4.24

Insight

Tracks monthly performance trends for executives.